

Only in this way can you systematically and successfully suppress EMI in an **HF** (RX/TX) Amateur Radio Station!

-A bit of EMI/EMC theory and optimized, example-driven implementation-

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<https://www.qrz.com/db/hb9cvq> <https://www.euro-emc-service.com>

HF = shortwave

Is a noise floor close to **S9 ->160m** / **S8->80m** / **S7->40m** in **SSB** (or even higher) **normal**?

RX S9= $50\mu\text{V}/50\Omega = -73\text{dBm} = 50\text{pW}$
TX= $10\text{W} = +40\text{dBm}$, $100\text{W} = +50\text{dBm}$, $1\text{kW} = +60\text{dBm}$

Definitions:

- + **EMV**: Elektromagnetische Verträglichkeit in der Elektronik [**DE**] **Devices and Systems**
EMV: Electromagnetic *Vulnerability* [**EN**] Susceptibility in Electronics, Boxes and Systems
- + **EMC**: Electromagnetic Compatibility in Electronics Equipment and Systems [**EN**]
- + **EMI**: Electromagnetic Interference in Electronics [**EN**]

Prepared in [EN] additionally to [DE] version for HAM-RADIO, 27-Juni-2026, FN

Short Bio:

Diethard E. A. Hansen (Ph.D.) MS/BS EE

All degrees from Germany: **Dr.-Ing.**, Dipl.-Ing. TU + FH

IEEE **Life Member** (USA), IEEE EMC-Society **Senior Member**,

Senior iNarte EMC Eng., **Radio Licenses** FCC: **AK4IG** 2011, CEPT: **DK2VQ 1968, HB9CVQ 1984**



- Since **1983** **professional** experience (**private Consultant**/registered EU-Auditor/iNARTE 34Y) in **Electromagnetic (EM) Disturbance and Interference Control in Electronics**, incl. **commercial/ government/ military** installations/equipment
- 160+ international technical **papers**/ 50+ **patents** are assigned to him.
- **Assessments/Audits**: 400+ Test-Labs (worldwide), incl. EU-D EMC, RTTE/RED and Automotive EMC
- Since over 30 years active in international **EM standards/regulations**. IEEE **Senior EMC Life Member** since 2020
- **1983 till 1991** **EMC Lab-Manager** in **R&D** for major, international electro tech. **Industry Company**
- 1991 to 2000 **private** Test-Lab/Certification Body **Owner**
Berlin Teltow was an accredited EMC Lab/competent Body EMC
Today this would be a Notified Body under EU-CE.

Webinar 05/2024 Video: <https://www.euro-emc-service.com/en/EMC-Radio-interference.htm> [EN]

40+Years in EMC and related Areas- Lessons Learned! -

HAM-Radio Meeting Munich, March 2023: https://www.youtube.com/embed/2J_wSaWFUNA [DE]

7+ Years (part time) HF-mobile Antenna R&D

Topics:

1. Why does **EMC** (electromagnetic compatibility) keep getting more important?
2. The only method to use: the **EMC Protection Zoning Concept**
3. On HF always **mitigate conducted EMI first**, then radiated!
4. EMC planning/retrofit, proper **EMC design** of an HF (RX/TX) amateur radio station
5. **EMC case study** HB9CVQ, Berikon/AG, on the 3. floor of a **6-party rental house** - no problem!
 - Measurements: Emission (**RX** noise floor) -- station off — in the shack and within the house
 - Measurements: Emission (**RX** interference environment) — station on — to dummy load — in the shack (house ⇒ zero CVQ-EMI)
 - Measurements: Emission (**TX** interference coupling via dummy load) — at the operating desk (house ⇒ almost zero CVQ-EMI)
 - Measurements: Emission (**TX** interference coupling via doublet dipole (160–10 m) antenna — station/230 VAC house wiring)
 - Experience: Emission (**RX** local QRM) — local QRM eliminator DXE NCC-1 (primary/aux antenna arsenal / RX protection)
6. **EMC case study** Multi-OP **Contest-Station DD1A**, Singen
7. **EMC case study** HF-mobile, Audi A6, **DK2VQ/m** operating experience
8. Summary and optimized **practical recommendations**

1. Why is EMC getting more important? (1)

- QTH EM-noise-environment/noise-floor; communications S/N ratio
- CE/China: device engineering and standards
- Terminology definitions: EMV/EMC/EMI (Electromagnetic Interference), RFI, EM noise (ITU)
- Emission (general = disturbance; in a telecom channel = interference), Immunity
- Time/Frequency domain
- Conducted vs. Radiated EM phenomena
- Wire-Line length vs. Antenna Efficiency; Near/Far field

As of summer 2020, VY2ZM operates the following antennas (excerpt): 160 m: 4-el full-sized phased BS/EF array; single 160 m vertical; 80 m: 4-square vertical array; 40 m: multiple 2-3 el yagis; 20/15/10 m: various; beverages; short vertical RX array, etc.

The Low-EMI: Dream of HF-Noise Silence

The King of 160m
K1ZM/VY2ZM



1. Why is EMC getting more important? (2)

- **Poorly** or illegally not **EMI-suppressed electronic devices**/installations are a problem almost everywhere today.
- They act as ‘unlicensed’ transmitters to HAMs, especially on HF.
- The result at the receive location (often in the near field) is a **reduced signal-to-noise ratio**.
- Frequently this **local QRM** appears as a broadband noise carpet or narrowband periodic interference (e.g., VDSL 4.3 kHz). PLC should have a 30 dB notch for HF amateur bands.
- The **VY2ZM** photo (Canada) shows an **ideal radio location** near salt water and virtually undeveloped surroundings.

1. Why is EMC getting more important? (3)

- Not everyone has a **W3LPL Super Contest HF-Station**



As of summer 2020, **VY2ZM** operates using the following complement of antennas:

160 - 4el full-sized phased BS/EF Dream Array designed by Peter Hutter and Rob Flory

- Single 160M Vertical (Omni Pattern)

80 - 4 square vertical array producing 5.68 db gain in 4 patterns

- 2el Vertical Array NE/SW

40 - 2el/2el 40m Yagi at 160/80 feet

- 2el 40M Yagi at 130 feet

- 3el 40M Yagi at 85 feet +++ wow 20/15/10m

Various **Beverages** at 880 and 1080 feet
Rx Sixpack of 6 short verticals for 160M NE/SW

1. Why is EMC getting more important? (4)

- At W3LPL (MD, USA), there is sparse development and an impressive, cleverly laid-out antenna farm (photo).
- For comparison, the vast 'antenna farm' of VY2ZM (Canada) is listed.

1. Why EMC ? Physics and important EMC Background Info (5)

Equations



James Clerk Maxwell (1831–1879)

$rot(\vec{E}) = -\mu \frac{\partial \vec{H}}{\partial t}$	Faraday's induction law
$rot(\vec{H}) = \varepsilon \frac{\partial \vec{E}}{\partial t} + \vec{j}$	Ampere-Maxwell law
$\varepsilon \cdot div(\vec{E}) = \rho$	Electrostatic law
$div(\vec{H}) = 0$	Equation of magnetism

Induktions-Gesetz

Durchflutungs-Gesetz

Gaussches-Gesetz

Gaussches-Gesetz des Magnetismus

Time dependence

Dimensions of a conductor vs. Wavelength determines basically it's Antenna Radiation Efficiency

Solution

Retarded magnetic vector potential

currents

distances

$$\vec{A} = \frac{\mu}{4\pi} \iiint \frac{\vec{j}(\vec{r}')}{|\vec{r}' - \vec{r}|} d\tau$$

Currents ↔ EM-Fields

$$\left\{ \begin{array}{l} \vec{H} = \frac{rot(\vec{A})}{\mu} \\ \vec{E} = \frac{1}{\varepsilon} \int_t (rot(\vec{H}) - \vec{j}) dt \end{array} \right.$$

uncontrolled EM Coupling/Radiation !?

Original Source: Frederic Pythoud_METAS_EMV_19.01.2017, SwissT.meeting , Fachtagung EMV Zürich Switzerland

1. Why is EMC getting more important? (6)

Maxwell's equations, simplified for HF amateurs: **alternating currents generate alternating fields**; such **fields induce currents in conductive structures**. Uncontrolled in/out coupling must be prevented by appropriate suppression measures/layout.

With electromagnetic alternating fields (unlike electrostatics) the **E (electric)** and **H (magnetic)** fields are inseparably linked.

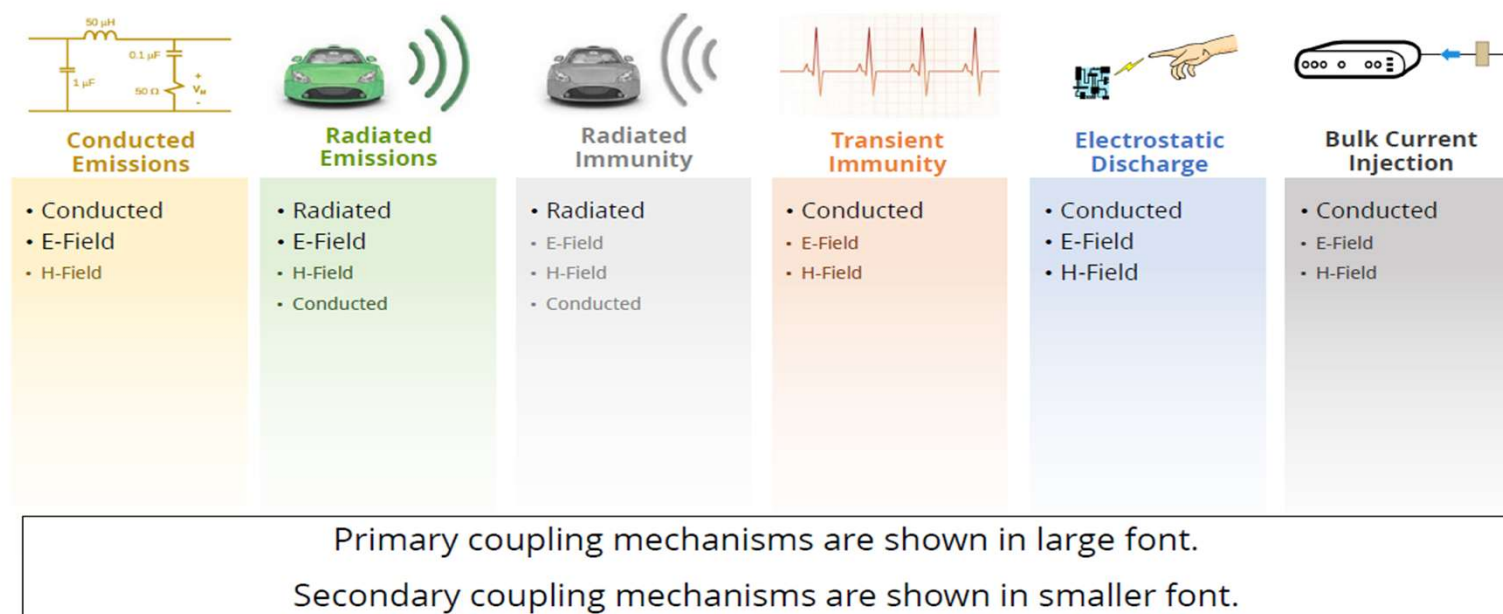
In the **near field** (e.g., magnetic loop vs. dipole/vertical) the radiator type determines the E/H ratio:
magnetic loop → **low-impedance/inductive**;

dipole → **high-impedance/capacitive**.

In the **far field**, **E/H** is distance independent at **377 Ω** with no phase shift (real power only) between E and H.

1. Why EMC ? Physics and important EMC Background Info (7)

EMC Measurements and Key Coupling Mechanisms



Source: Prof. Dr. Todd Hubing, LEARN EMC, IEEE EMC Symposium RENO, NV/US, 2020

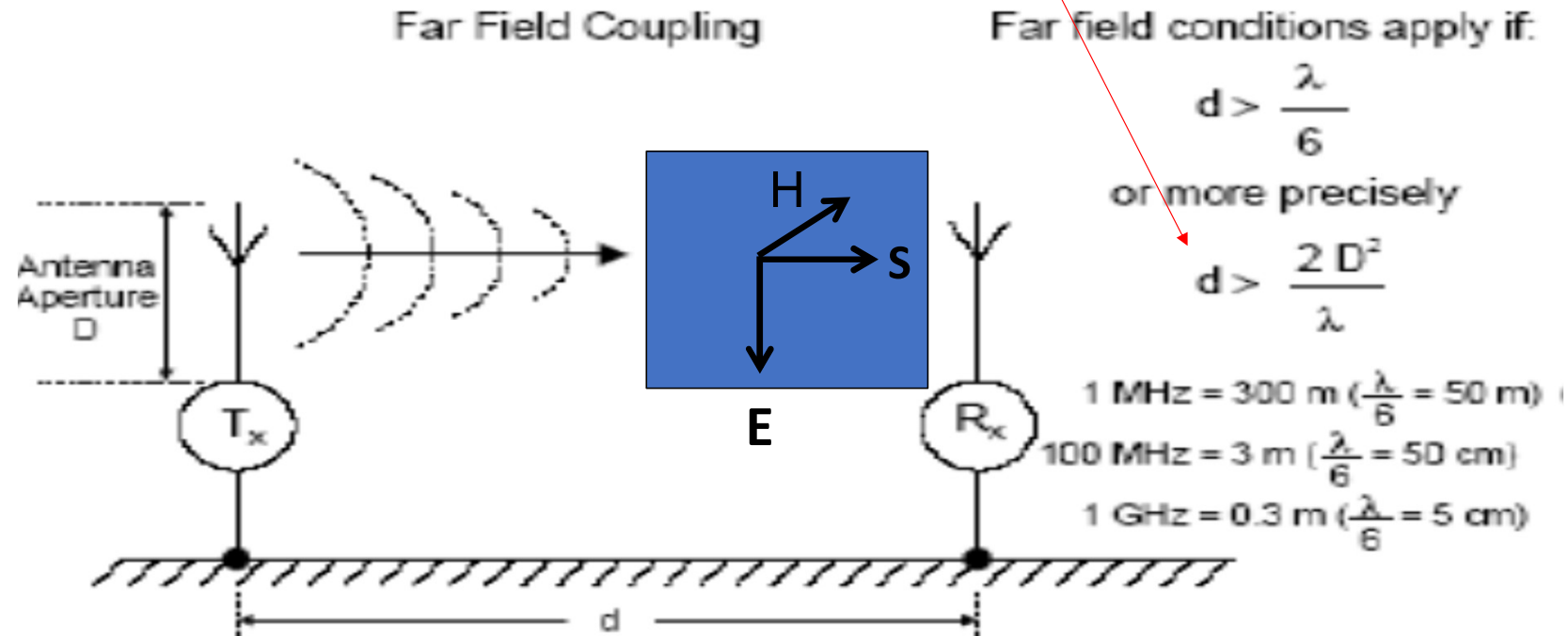
1. Why is EMC getting more important? (8)

- We distinguish **conducted** and **radiated Coupling Effects**.
- Both can cause **Emission** or **Immunity** (susceptibility) issues.
- We also consider **Frequency- and Time-domain** descriptions.

1. Why EMC EM-Field Basics (9)

Near Field / Far Field (FF)

More precise, better for FF (Far-Field) check



1. Why is EMC getting more important? (10)

- Depending on TX/RX distance, frequency (f [Hz] = c [m/s], wavelength [m]) and antenna size D [m], conditions for transition from near to far field are shown.
- In far field: $E/H = 377 \Omega$; vectors are orthogonal to propagation; S is the Poynting vector ($S = E \times H$) and denotes radiated power density.
- Examples: **1 MHz/300 m (MW); 100 MHz/3 m (FM Broadcast); 1 GHz/0.3 m (Microwave e.g., 2.4 GHz Mw-Oven, Bluetooth, WLAN...).**

1. Why EMC Basics (11)

Near Field / Far Field

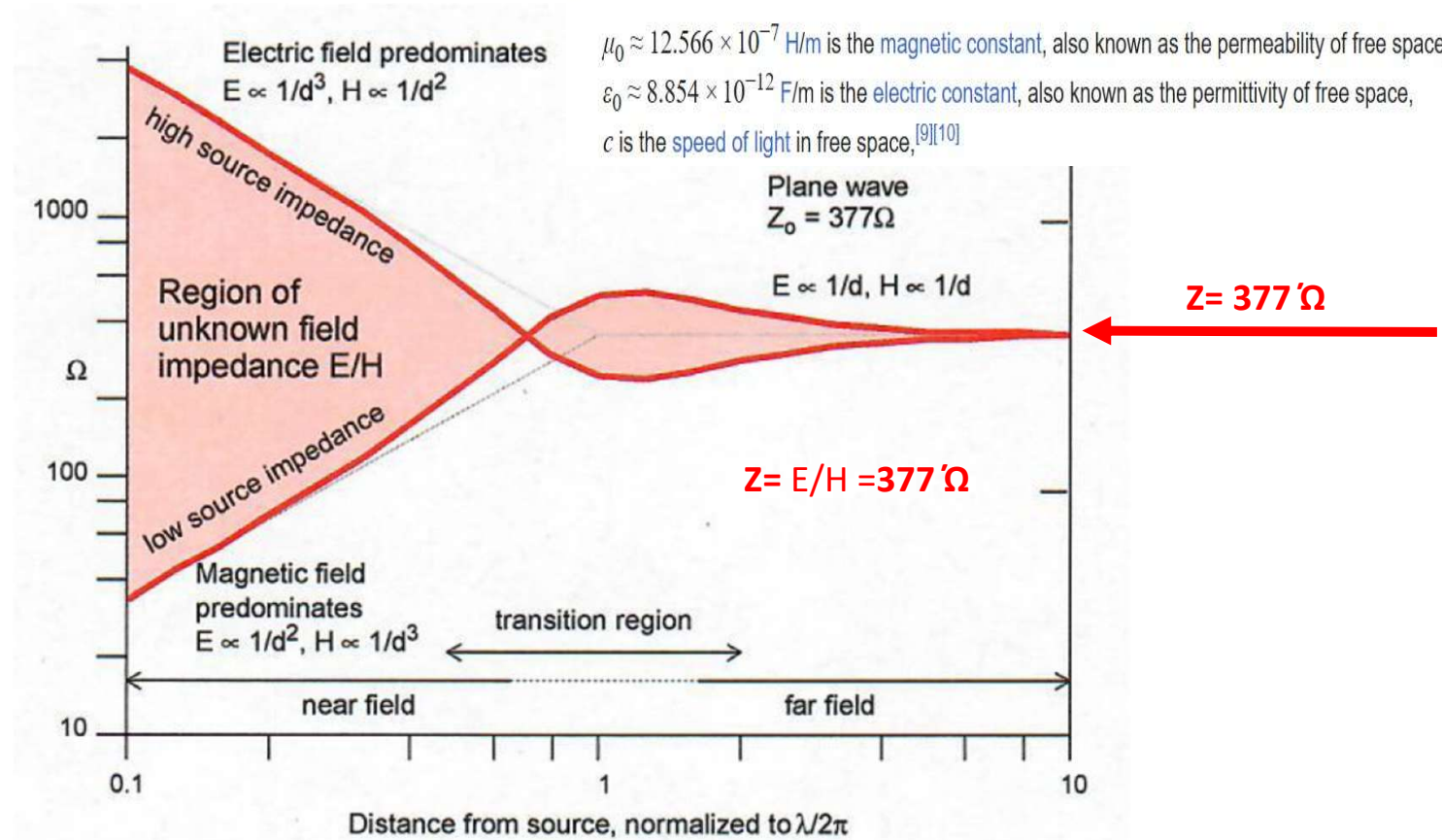
Dipole/Monopole
Antenna

Magnetic Loop
Antenna

$$Z_0 = \frac{|E|}{|H|} = \mu_0 c = \sqrt{\frac{\mu_0}{\epsilon_0}} = \frac{1}{\epsilon_0 c},$$

where

$\mu_0 \approx 12.566 \times 10^{-7} \text{ H/m}$ is the **magnetic constant**, also known as the permeability of free space,
 $\epsilon_0 \approx 8.854 \times 10^{-12} \text{ F/m}$ is the **electric constant**, also known as the permittivity of free space,
 c is the **speed of light** in free space,^{[9][10]}



Source: Schaffner Guide 2001

1. Why is EMC getting more important? (12)

- The graphic computes **free-space wave impedance ($Z = E/H$)** from physical vacuum constants.
- With increasing distance from the radiator (dipole/mag loop), the **E/H ratio approaches 377Ω** .
- Distance normalized by simplest near/far field estimate ($\lambda/2\pi$).

1. Why EMC Basics (13)

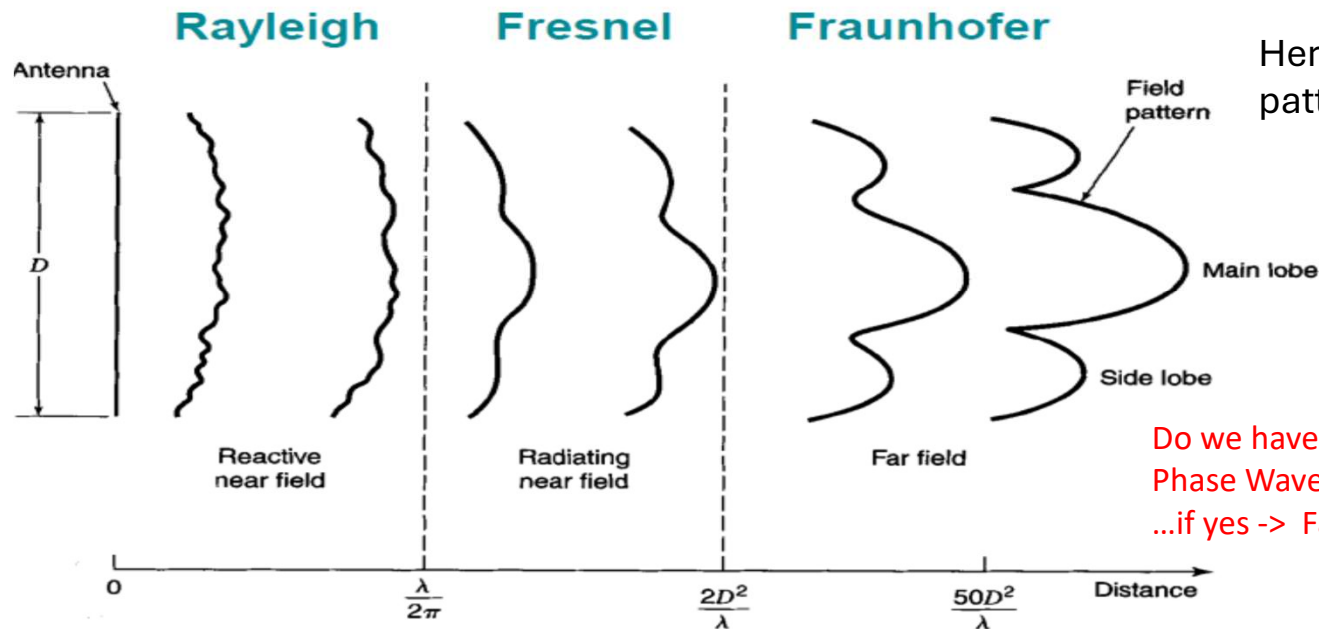
Near Field / Far Field

$\lambda/2$ Dipole (a Low-Gain Ant., not a High-Gain Parabolic-Dish Ant.) in ca. **2 bis 4λ** we reach FF

e.g., on 80m -> over 160m distance needed for a kind of FF-conditions !

Near Field (NF)

Far Field (FF)



Here the full radiation pattern is visible.

Do we have a (nearly) Planar Phase Wave-Front->**TEM** Wave ?
...if yes -> Far Field

$$Z = E/H = 377 \Omega$$

https://www.tu-ilmenau.de/fileadmin/media/it_hmt/vorlesungen/antennen/antennen.pdf Antennen, Prof. Dr. M. Hein, Summer Lecture 2016

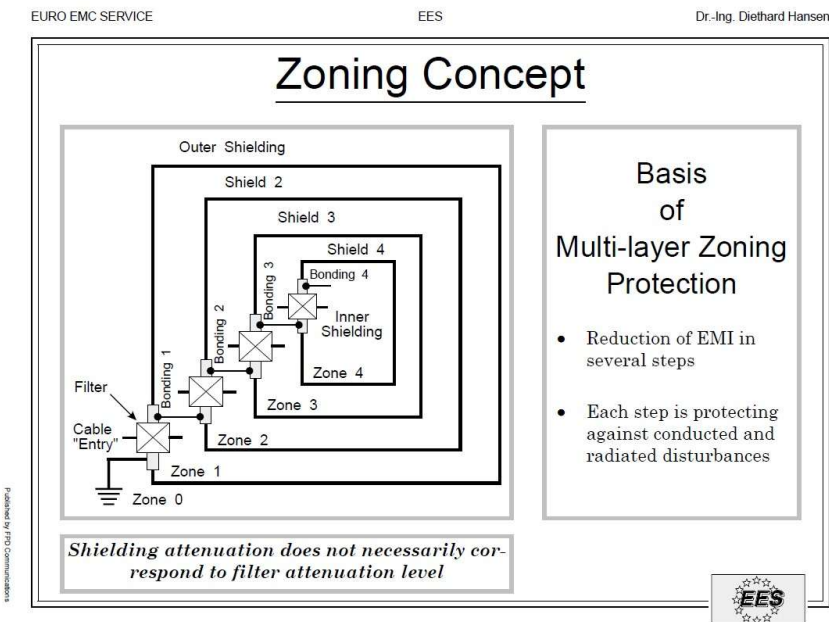
1. Why is EMC getting more important? (14)

- As one moves away from an antenna, **different zones are traversed until the true far field** is reached.
- For a **simple half-wave dipole** in free space, that's **$\sim 2-4 \lambda$** .
- For high-gain dishes, the distance can be much larger before the radiation pattern is 'fully formed'.
- The exact point depends on the **allowed deviation (e.g., 1 dB $\approx$ 11%)** from the ideal.
- In **true far field** we have a planar TEM wave; near the source, spherical propagation transitions to an almost planar wavefront at large distances.

2. The only method to use: the EMC protection zone concept (1)

Zone concept / topology concept / 'onion-skin' principle

On HF possibly also (semi-) open zone concept → reference => ground plane

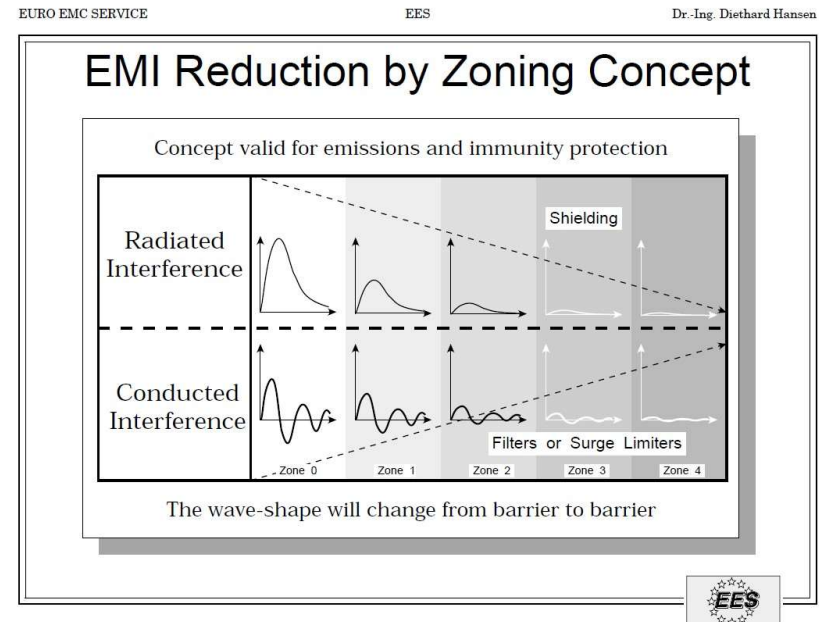


EES/SYS/17

At zone transitions, implement protection:

For external EM disturbances

on lines: via filters/limiters; **for fields:** via staged shielding



EES/SYS/19

EM-Threat from outside the system:

Step-by-step EM disturbances Reduction

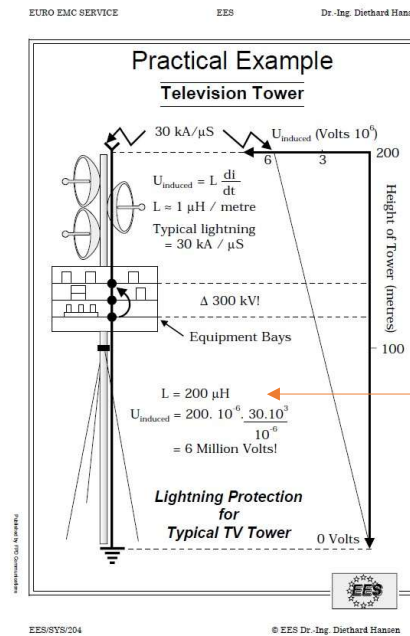
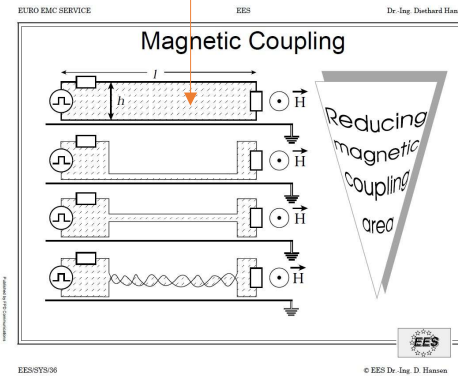
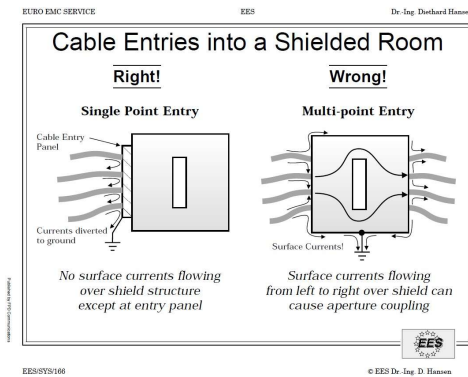
D. Hansen

2. The only method to use: the EMC protection zone concept (2)

- The EMC protection zone concept divides a sensitive electronic system into areas with different protection (currents/voltages/E,H-fields) requirements (e.g., **20 dB (1:10)**, 40 dB (1:100), **60 dB (1:1000)**, 80 dB (1:10,000) and higher), for disturbances from **outside (immunity)** and **inside (emission)**.
- **Conducted disturbances** are reduced with filters/surge protection; field-related disturbances with shielding.
- **Coupling mechanisms differ physically**: long cables have low resonances; smaller shielded volumes have higher ones.
- **Incident waves experience**: reflection, some absorption, and transmission depending on frequency, material, conductivity.
- Ideally, each zone has only one physical entry/exit (**single-point entry**), otherwise uncontrolled shield currents can flow and leak fields via unavoidable apertures (e.g., ventilation).

2. The only method to use: The EMC protection zone concept (3)

- Single-Point Entry (control disturbance current paths by design)
- Nearby HF reference ground planes
- EMC installation impedance isolation; “HF isolated island” concept
- Filters/shields work by insertion loss (with/without)
- Avoid/minimize critical (RX/TX) coupling loops
- Coordinate with lightning protection (e.g., TV tower).



2. The only method to use: the EMC protection zone concept (4)

- By **Single-point entry** disturbance currents will be channelized.
- Introducing **HF reference ground planes minimizes loop areas** and prevents/minimizes unintended antenna structures — especially for sensitive signaling.
- **Filters are defined by their insertion loss** (with vs. without filter, in linear ratio or dB).
- Long ground/earth leads >> inductance ($\sim 1 \text{ m} \approx 1 \mu\text{H}$) become ineffective even on lower HF bands
- (e.g., 40 m).
- The goal is to **isolate the shack from the outside world** as an **HF island** for 160–10 m.

2. The only method to use: the EMC protection zone concept (5)

For **lightning protection** of e.g., a 200 m TV tower with a restaurant, distinguish lightning protection (people/fire/property) from **EMC protection** (disturbance/destruction of electronics).

- A massive down-conductor must safely carry **tens of kA to ground**. Lightning **spectra** are **mainly below 1 MHz**;
- **Inductive reactance** is still present and causes **huge kV drops (top-to-bottom)**.
- The lightning protection concept (**transient protection, bonding**) controls dangerous voltages to prevent arcing e.g., between floors/levels.
- Lightning bonding aims at **potential equalization with far lower HF demands than EMC** (which needs meshed ground grids with small cell size to minimize inductance).
- **Both protection concepts must be coordinated** in practice; **always minimize loops**.

3. On HF: mitigate conducted first, then radiated! (1)

- Transmission-Line/Antenna theory, myths vs. facts on efficient grounding/bonding ($V_{ind} = L \cdot di/dt$) => avoid unwanted HF voltage differences between equipment
- Real-world effectiveness of filters/shielding in systems (filter test impedances vs. real installation impedance !)
- Proper clamp/core ferrite selection; frequency-dependent insertion impedance (choke efficiency); current-saturation effects
- **Electrical safety earth (PE) or lightning protection is unsuitable for HF!** ($V_{ind} = L \cdot di/dt$).
Rule of Thumb: 1 m of wire $\approx 1 \mu H \rightarrow X_L = 2\pi f \cdot L \Rightarrow$ **at 7 MHz, 1 m $\approx 44 \Omega$; at 14 MHz $\approx 88 \Omega$...**
- Cables behave as **transmission lines** (telegrapher's equations). For height above ground $h > \sim 0.1 \lambda$, simple theory breaks down
- The **conductor becomes an antenna**: from $\sim \lambda/10$ (sometimes $\lambda/100$), surely if 1/2-wave dipole resonance, 1/4-wave GP resonance, etc.
- Vertical Rods act as monopoles; Loops as loop antennas.

Student HV-lab experiment: Copper floor with spark gap triggered by 8/20 μs surge currents — illustrates $V_{ind} = L \cdot di/dt$.

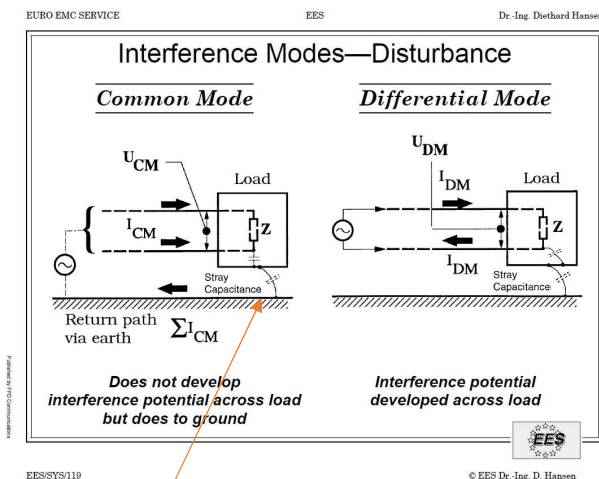
Driving a short lightning test pulse current over a metal plane will trigger an open spark gap (point to point) soldered onto that ground plane due to finite inductance of the metal floor.

3. On HF: Mitigate conducted first, then radiated! (2)

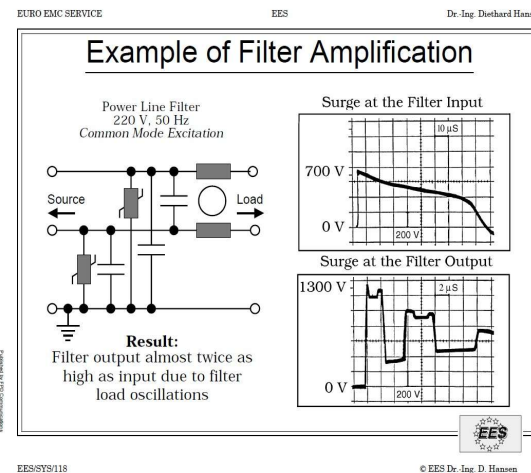
- Lightning protection goals differ from EMC/EMI goals.
- Physically, first separate line vs. field (antenna) theory.
- A conductor becomes an **efficient radiator from $\sim\lambda/10$** (sometimes $\lambda/100$).
- The induced voltage drop $V_{ind} = L \cdot di/dt \rightarrow$ inductance of the ground lead is decisive: 1 m $\approx 1 \mu\text{H}$ ($X_L \approx 44 \Omega$ at 7 MHz; 88Ω at 14 MHz). Only by creating wide-area meshes can inductance and thus voltage drops be lowered.
- But if the area approaches $\sim 1/10$ of antenna dimensions, equipotential equalization no longer holds; Standing waves form and the protection concept collapses.
- Remedy: create smaller ground 'islands' and integrate smartly.
- **Do not blindly trust catalog filter/shield attenuation numbers**; they are lab values under standard impedances and only partially relevant in systems (e.g., mains filters see different impedances/wiring).
- For shielding, distinguish **near/far-field tests** by standard vs. by application.

4. EMC planning/retrofit, proper **EMC design of an HF (RX/TX) station (1)**

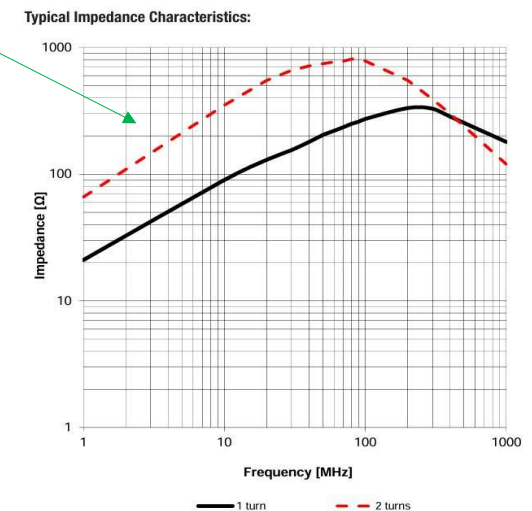
- HF equipotential bonding in the station: TRX, PA; thin aluminum ground plate under the operating desk;
decouple PE/safety ground/earth for HF!
- Common-mode (CM) vs. differential-mode (DM) currents
- CM-Chokes: Mains filters; computer/USB ferrites (raise shield L/R $\rightarrow$ increase Z); rotor control; coax etc...



At HF CM is basically always dominant



Filters in Systems can cause adverse effects
e.g., resonances



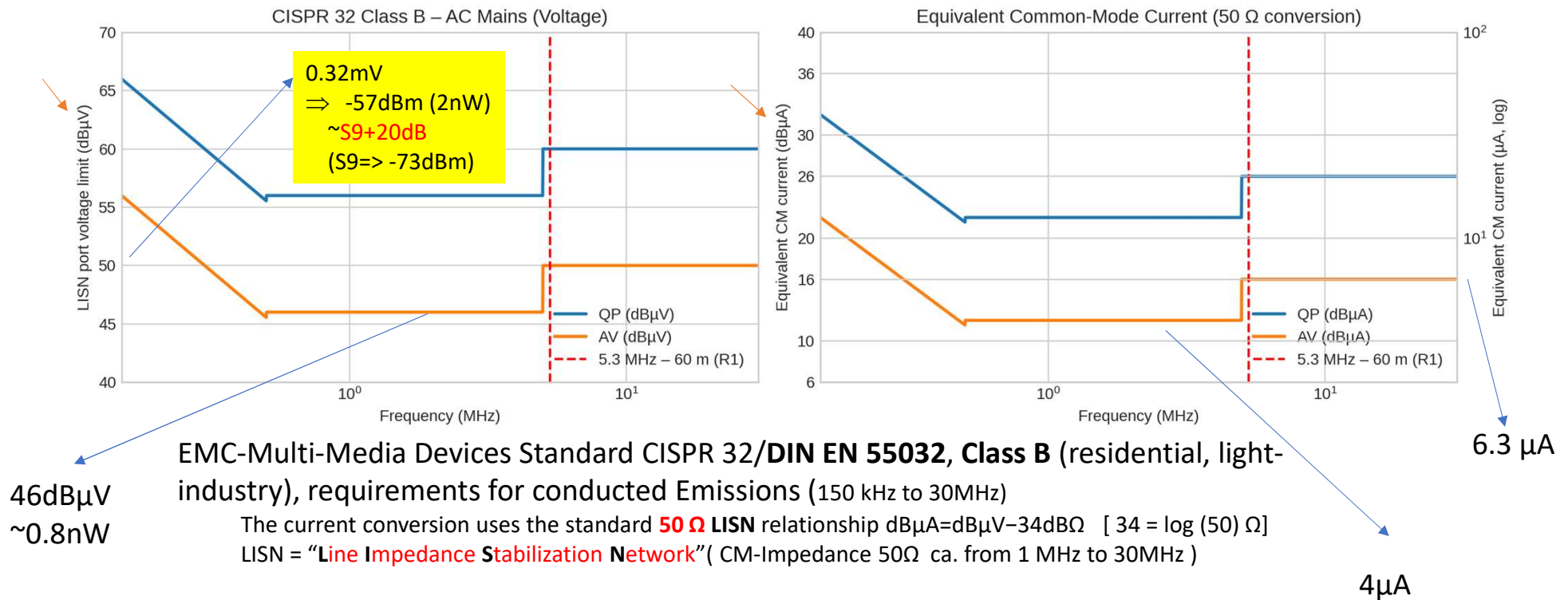
Würth 74271722S (Core is for 1.8 MHz ok)
Snap on Ferrite, 620 μ r D Kabel max. 12.5 mm

4. EMC Planning/Retrofit, proper **EMC Design of an HF (RX/TX) station** (2)

- Mains filters suppress **CM and DM** disturbances.
- **Pay special attention to filtering of PE**; otherwise, large coupling **loops can form via cable shields (ground plate/single-point entry) to antennas — bad for both TX and RX.**
- In **real filter installations, length/load-dependent resonances ($Q \sim 3-5$)** can occur in frequency domain; in **time domain**, max **factor is 2** (e.g., total voltage reflection at open).
- With **snap-on ferrites, compress the air gap** with a cable tie to maintain long-term performance.
- With **high-permeability cores**, use **no more than 3–4 turns** (parasitic C); that already yields kilo-ohm impedances for CM choking.
- **Ferrites work at current** antinodes (**low impedance** — verify with current clamp/H-field probe), not at voltage antinodes along a line.

4. EMC Planning/Retrofit, proper EMC Design of an HF (RX/TX) station (3)

AC Mains Conducted Emission Limits – CISPR 32 Class B
Continuous Curves with 5.3 MHz (60 m, Region 1)



4. EMC Planning/Retrofit, proper EMC Design of an HF (RX/TX) station (4)

On HF, CM almost always dominates. HF CM current flows on the outside of coax. A ferrite on the outside does not affect signal inside; added outside impedance **reduces** uncontrolled loop currents.

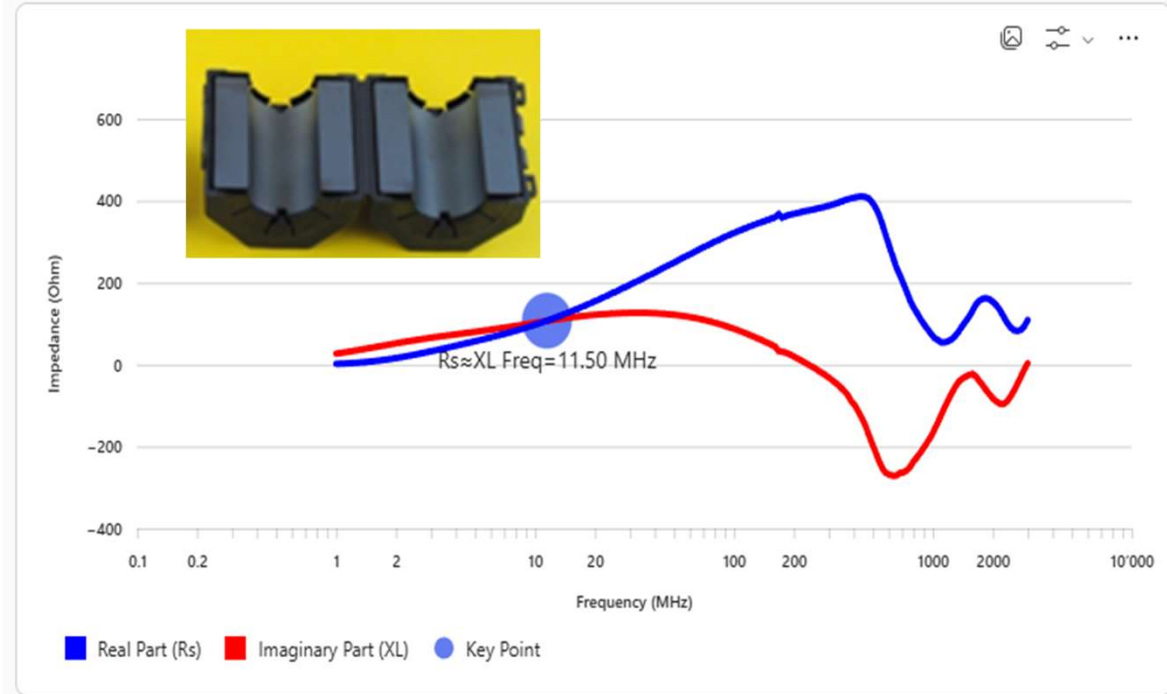
- **Ferrite** (Shield-L,R-> increases Z), choke off coax, control/rotor-cable....

Graph: 1 Turn/Winding (N=1),
300g heavy Snap-on Ferrite-
Core : **Fair-Rite Part**
#0444177081

Dinner ca. 25mm

Tip: with a single large snap core (Fair-Rite #0444177081, ~25 mm ID), 3–4 turns are often enough on HF to achieve ~20 dB ($\approx 1 \text{ k}\Omega$) CM reduction.

1. Magnitude of Impedance $|Z|$ vs Frequency (MHz)



4. EMC Planning/Retrofit, proper **EMC Design of an HF (RX/TX) station** (4a)

- How much CM (choke R_c) suppression (dB, I with / I without choke) do you need at the antenna (Z_a)?

Why the design target is “big Z choke” (2–5 k Ω resistive)

Treat the choke as a series element in the CM loop. If the antenna-side CM impedance is Z_a and the choke is mostly resistive R_c , the **current ratio** is roughly:

$$\left| \frac{I}{I_0} \right| \approx \frac{|Z_a|}{|R_c + Z_a|}$$

- With $R_c = 3 \text{ k}\Omega$, $Z_a = 50 \Omega \rightarrow -35 \text{ dB}$.
- With $R_c = 3 \text{ k}\Omega$, $Z_a = 200 \Omega \rightarrow -24 \text{ dB}$.
- With $R_c = 3 \text{ k}\Omega$, $Z_a = 1 \text{ k}\Omega \rightarrow -9.5 \text{ dB}$ additional beyond the first 20 dB (i.e., even better).

RA=200 Ω
RC= 1000 Ω
=> -16 dB

This math is why **K9YC** and **ARRL** guidance converges on **multi-k Ω , largely resistive** chokes for HF—it keeps CM current small even when the antenna-side CM load is inconveniently low or reactive. Measured Data For Ch... +1

4. EMC Planning/Retrofit, proper **EMC Design of an HF (RX/TX) station** (5)

Why ferrite effectiveness varies

- + Ferrite effectiveness in a system is frequency- and location-dependent
- + Current/voltage distributions (standing waves) exist along cables.
- + A ferrite (modeled as series L+R with some stray C) works well at current antinodes (low Z).
- + Measuring with a **50 Ω VNA** or **HF-impedance meter** is useful for material orientation but can be **tricky** at very low/high impedances (de-embedding and fixture stray C).
- + In systems, **it's better to measure CM current insertion loss on-site with an HF current clamp.**
- ++ Example:
 1. replace a 60 cm 'choke section' with plain coax for 0 dB reference,
 2. Then insert the choke section and measure current reduction in dB.
 3. **Insertion loss [dB] = $20 \cdot \log_{10}(I_{CM,ref} / I_{CM,choked})$.**
- + Where to get an **HF current clamp**? Buy (pro: >€500) or **DIY** (<€50)
- ++ Transformer principle: Primary 1 turn; Secondary N (e.g., 6–10 turns) with ~50 Ω load;
Measure with spectrum analyzer (use attenuator; beware TinySA max +6 dBm).

4. EMC Planning/Retrofit, proper **EMC Design of an HF (RX/TX) station (6)**

DIY current probe: https://www.interferencetechnology.com/wp-content/uploads/2012/04/Wyatt_NA_DD12.pdf

Trafo Prinzip => Primary (1N) : secondary N (6 to 10) und parallel e.g. 50Ohm Load-R) 1N

Commercial EMC monitoring Current Probe

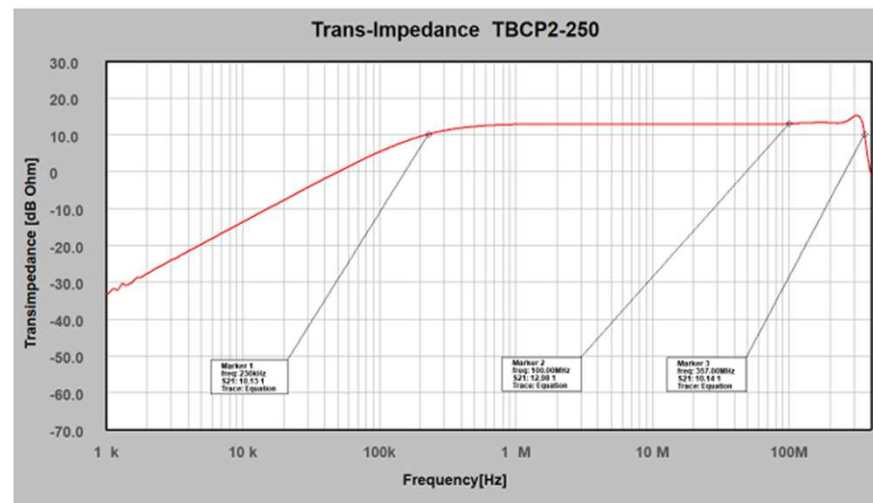


Figure1: typical transfer impedance: 1 kHz to 400 MHz



Ohm's Law $I=V/R$ linking [μV , μA]

=> In Log System:

$dB\mu A = dB\mu V - dB\Omega$

$[dB] = 20 \times \log (I_{CM \text{ no Ferrite}} / I_{CM \text{ with Ferrite}})$

Ratio 1:10=> 20dB

4. EMC Planning/Retrofit, proper **EMC Design of an HF (RX/TX) station** (6a)

Measurement caution with TinySA (Spectrum Analyzer battery operated)

Quick reference: DIY probe math (N=10 turns, R_b=50 Ω)

- $I_{CM} = N \cdot V / R_b = 10 \cdot V / 50 = 0.2 \text{ A/V}$
- 1V → 200 mA
- 100 mV → 20 mA
- 50 mV → 10 mA
- 20 mV → 4 mA

Improvement (dB) = $20 \log_{10}(I_{\text{after}} / I_{\text{before}})$



TinySA max input $\approx +6 \text{ dBm}$ ($\approx 113 \text{ dB}\mu\text{V}$, $\approx 0.45 \text{ V}$ into 50Ω) — always use an attenuator (20dB+?) to **avoid damage**

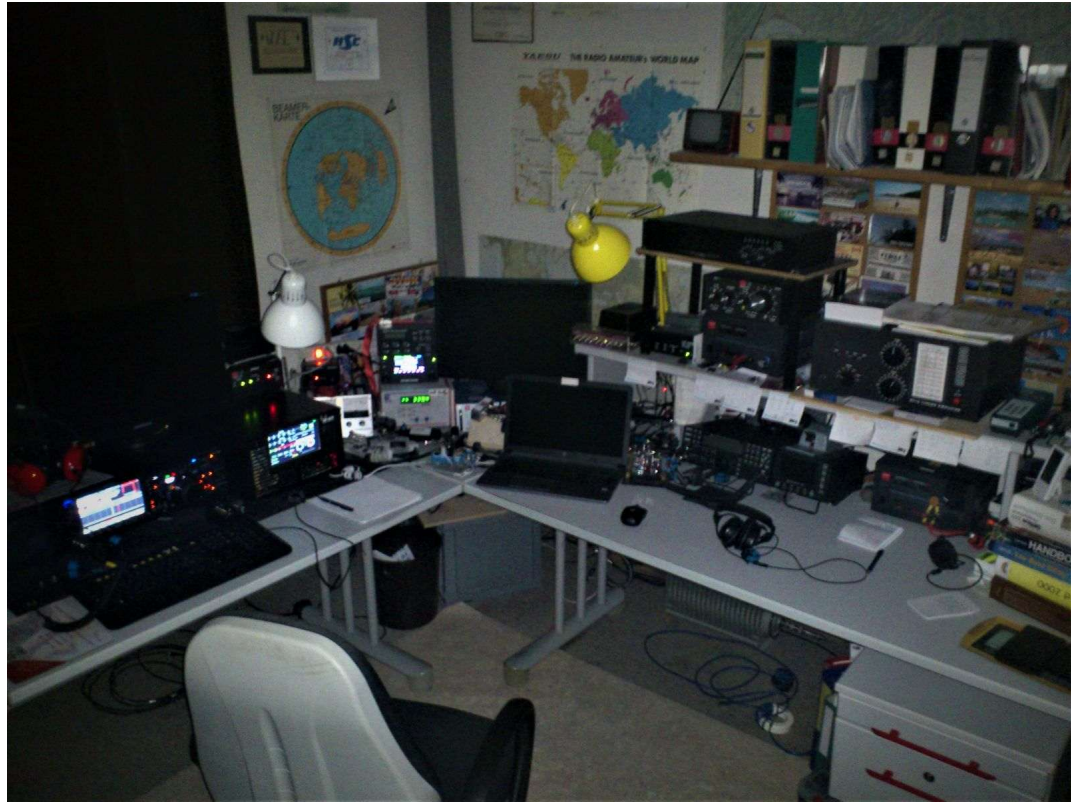
4. EMC Planning/Retrofit, proper EMC Design of an HF (RX/TX) station (7)

-Ground-plane HF-Reference and Cable Management-

- Use a sheet **ground-plane** (<1 mm Al; skin depth on 160 m $\approx$ 63 μ m) under the desk to minimize loop areas.
- Bond gear with **short studs** through the wood top to the plate.
- **All cables to/from the desk are choked off**: all coax, control lines , rotatory cable mains filters preferably with surge arresters etc.
- **On the desk**: TRX USB, PTT, CAT, headphones, mic, audio, external DC PS.
- Beware: some external 13.8 V supplies tie PE to DC-; TRX chassis may be at minus => then not all current returns via the black lead; some flows via PE; CM ring cores can saturate and lose effect.
- If the **TRX has an internal AC supply**, a CM choke on the 3-wire mains lead is fine.
- Warning: **unknown flea-market cores** should be e.g., VNA-pre-characterized
- **VNA test fixtures** can introduce some errors via stray C to the instrument front panel; build a well-defined **transmission-line jig** and calibrate properly.

5. EMC case study HB9CVQ — Berikon/AG — 3rd floor of 6-unit rental — no problem! (1)

DB18E SteppIR (40–10 m) @19 m; 2×36 m tuned doublet (160–10 m) @24 m. Single-op: FTDX101MP; RF2k-S or K3S; P3; KPA500/ACOM1k.



5. EMC case study HB9CVQ (2)

- Home QTH with antennas and shack (upper right).
- The installation is **an HF-isolated island with external lightning protection.**
- Internal lightning bonding in emergencies via a triggered spark gap (tested 100 kA) to a separate earth system.
- Outside: proper mast down-conductor to a regularly measured 14m deep earth.
- In the shack two phases are used (PA, TRX).
- K3S, P3, KPA500, ACOM 1K is reserve only
- No SO2R at present.

5. EMC case study HB9CVQ (3)



View above the beam
at 24 m: NW, S, E, SE
(photo).

5. EMC case study HB9CVQ (4)

- HF-capable bonding in station (TRX, PA), ground plate under wooden desk, short bonding studs;
 - CM-Choke in PE safety earth
 - good Mains filters; computer/USB ferrites; rotor control; coax
 - CM/DM currents well controlled
- 1) Discuss differing effects for emission vs. immunity.
 - 2) Semi-open zone concept on corner desk (OP1/OP2 positions) — has already a semi-critical resonance already above 21MHz.
 - 3) CM ferrites everywhere: mic, headphones, key, PTT, coax, mains.
 - 4) Two single-phase feeds (TRX, PA); mains filter cases lie flat on ground-plate (minimize L).
 - 5) Narrow cable channel to attic and roof antennas (with spark gap for lightning — otherwise this would form an EMI loop via coax shield to bonding, grounding, earthing).
 - 6) Next improvement a properly bonded metallic cable tray (HF-reference plane) from shack to attic/roof antenna distribution/installation (minimizing stray effects/loop areas/effective antenna height issues)

5. EMC case study HB9CVQ (5)

- The **Zoning Concept** can be implemented on HF even as a semi-open design with only a reference ground-plate (no full Faraday cage needed).
- On **receive** (**emission case**), noise sources (local QRM) are typically outside the shack/flat; induced currents are in the **μA range**.
- On **transmit** (**immunity case**), the antenna near field induces currents in the installation;
- The RX-oriented EMC measures also help here; currents scale with TX power and can quickly be **10mA to 100+mA range**.
- For **RX-noise compensation**, a DXE NCC-1 **two-channel phasing system** is used; physically, it can ideally cancel only one phase-aligned source (-180°) in the near field (local QRM Eliminator).
Practice shows compromises and success of ~3–4+ S-units (**≈18–24 dB**) local QRM suppression. In Lab/Theory suppression may exceed 40dB.

5. EMC case study HB9CVQ (6)

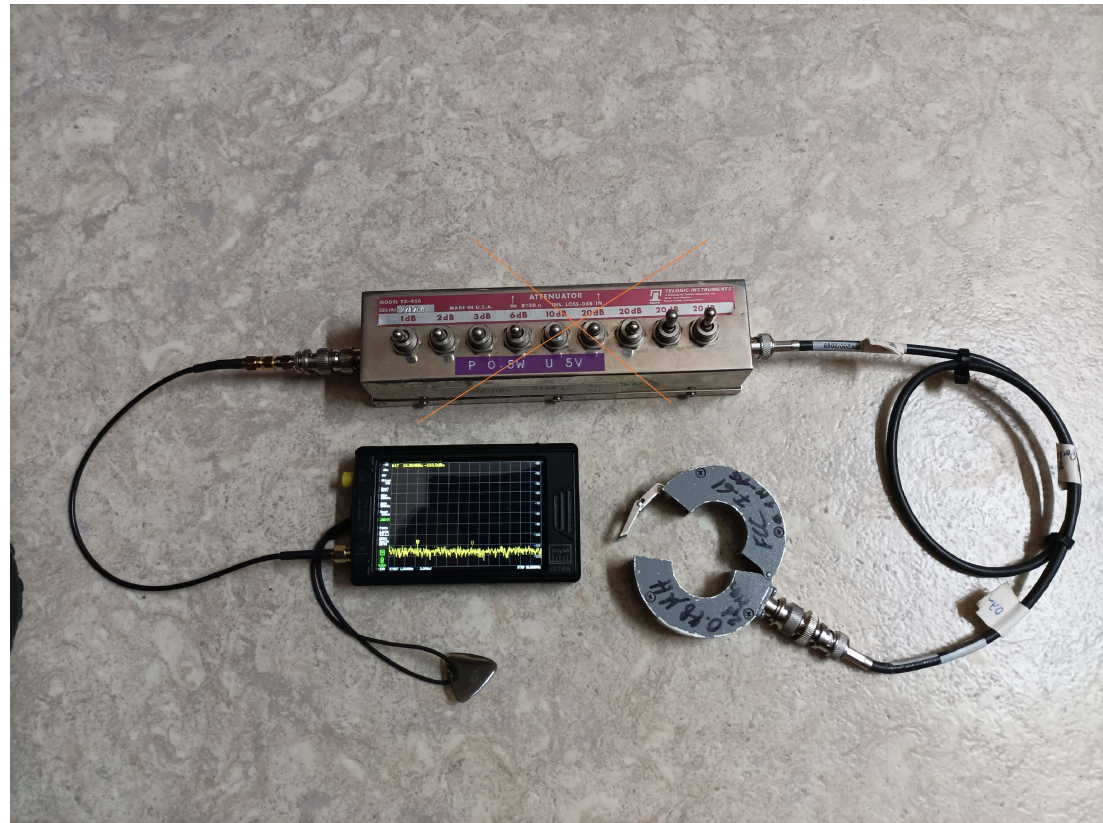
Practical setup for conducted emission/coupling measurements (see source photos/equipment list).

Always do a “Null Check”
(current probe closed, TX-field exposure) of the system not to suffer uncontrolled coupling.

Finding:

Metallic Attenuator Box presents too much surface and caused stray effects, field coupling into TinySA.

Solution: Use a small 2Watt BNC 20+ dB Pad. The TinySa ultra showed no problems even at 1kW TX power at maybe 30m distance to the 20m Beam antenna. (Alternative: Use snap-on ferrites CM cable choke)



5. EMC case study HB9CVQ (7)

The dBm Ladder according to W2VJN (2009), excellent info

https://www.vibroplex.com/techdocs/INRAD/MII_W2VJN.pdf

RX: S9 = -73 dBm

TX: 1 kW = +60 dBm (HB9 HF-max. legal limit mostly)

Dynamic range already here $73 + 60 = 133\text{dB}$!

An S-Unit below S9 (IARU=>6dB, Jap. Manufacturer =>3dB)

6dB=> 4xPower, HF S-Meter calibration => in RMS (S9: 50µV/500hm)

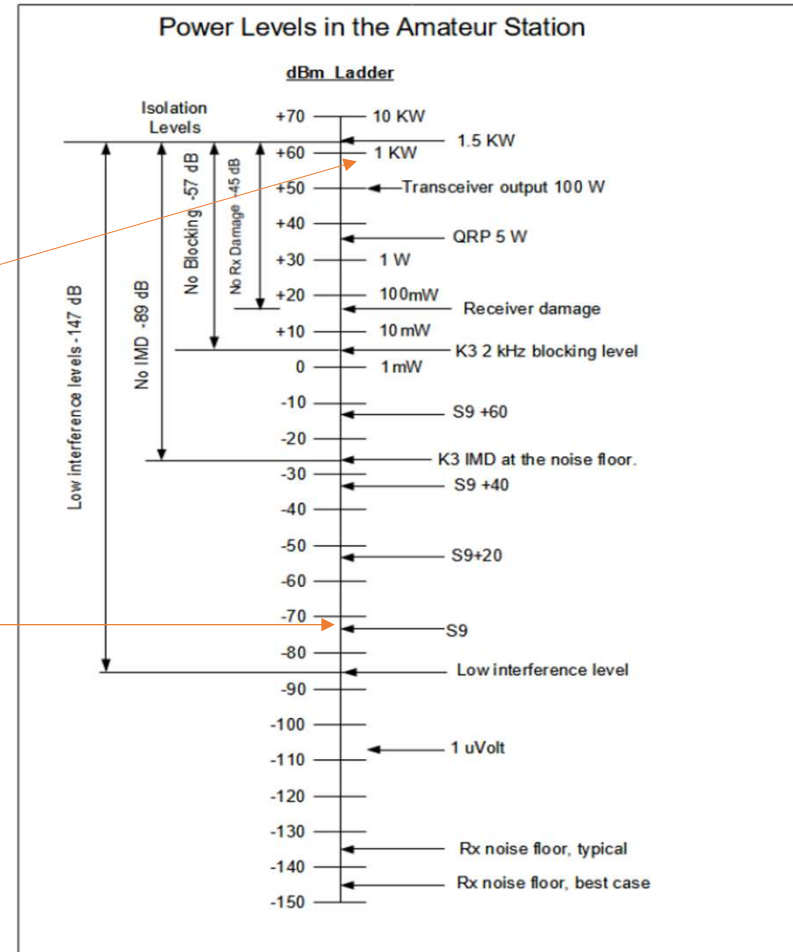


Figure 1 The dBm ladder.

5. EMC case study HB9CVQ (8):

How good is the EMI suppression?

Station desk: 1× antenna cable bundle to attic; 1× phase TRX; 1× phase PA; 1× earth lead with spark gap.

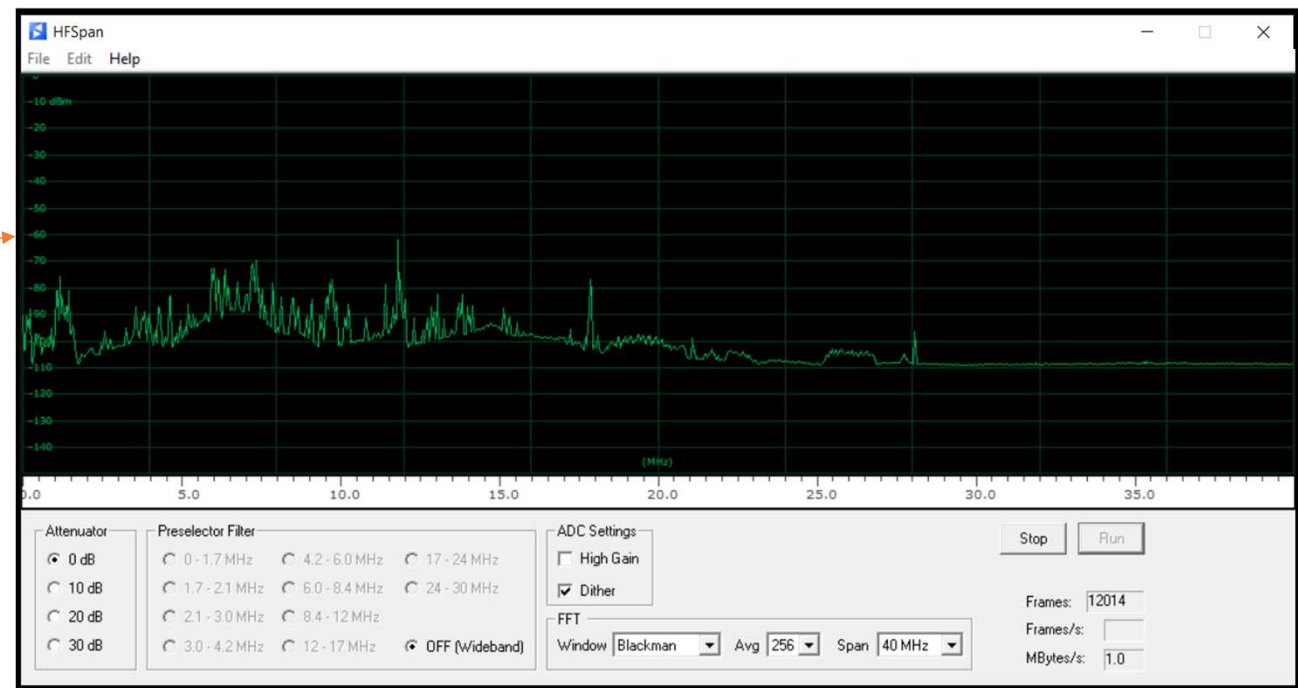
CM-current measurements — emission (RX noise floor) — **station OFF** — in shack

-60dBm/50Ω =>47dBμV
(12MHz, I-Probe 23dBΩ):
24dBμA=16μA (13nW)

11-Jan-2026, 17:52 MEZ
Per Current Probe FCC-61

Perseus in BB-Mode, Laptop, all
battery operated

All other currents are smaller



5. EMC case study HB9CVQ (9):

How good is the EMI suppression?

CM-Current Test:

1. Emission -Station ON-> into **Dummy Load** => TX-10W-CW → no relevant coupling , proving EMC concept is Ok
2. Emission -Station ON-> into **Dipole Ant.** => TX-10W-CW → the antenna illuminates the LV installation (attic first)

Induced CM power (dBm->calculated decoupling dB) on the **cable bundle to attic, referenced to TX 10 W (+40 dBm)**

shown is approx. **decoupling [dB] by band:**

160-80-60-40-30-20-17-15-12-10 m
62-60-60-50-52-45-50-**43**-60-50 dB

The bundle is about 30 m long ⇒ possible 2λ resonance around 15 m band (only **43 dB** decoupling there).
+40 dBm – 43 dB = –3 dBm (~0.5 mW, 50 Ω) induced ⇒ **~3 mA**. Scaled to **100 W** ⇒ **~9 mA**; to **1 kW** ⇒ **~30 mA**.

5. EMC case study HB9CVQ (10):

Local RX-QRM eliminator DXE NCC-1 (Main, Aux-Antenna Arsenal, RX-Input Protection)



Principle: improve **SNR** -truly works best for a single interference source. In the lab up to ~45 dB; in practice usually ~24 dB (≈ 4 S-units).

Procedure: optimize the noise-aux channel, invert by -180° , vector-sum via power combiner; match main-antenna and aux-antenna levels within ~1 dB while considering RX S/N. Often ~15 dB preamp is required for the aux ant.

Various small aux antennas were installed in the attic (short dipole; active mag loop ALA1530; Mini-Whip). Place ferrites along the coax to the aux antennas.

Sometimes, for **very close (~50 m) noise sources, a mains LISN** ($50\ \Omega$, $50\ \mu\text{H}$) with surge protection is **used**; it usually needs gain (~20 dB). Caution: LISN draws high 50 Hz leakage (e.g., 700 mA at 230 V/50 Hz) $\rightarrow$ RCD (30 mA) will trip. **Using the mains as a (poor) receive antenna ($\approx -25\ \text{dBi}$)** aggregates multiple sources with random phasing. Attempts to reduce far-field OTH radar on 17 m achieved ~ -20 dB but required constant phase readjustment due to ionospheric variability — better done by software/automatic diversity RX.

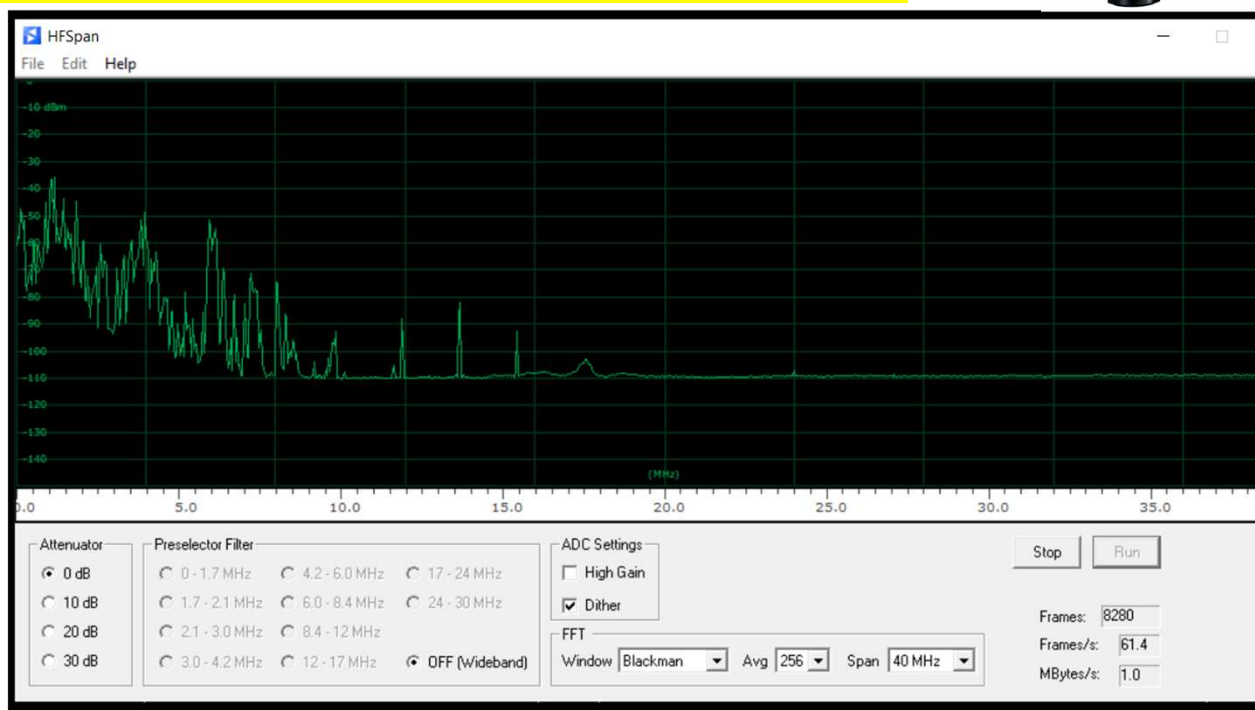
5. EMC case study HB9CVQ (11)

Signal level examples (Main antenna Dipole)

14-Jan-2026-2000CET-1872 kHz (160m) tuned «inverted Vee» @24m



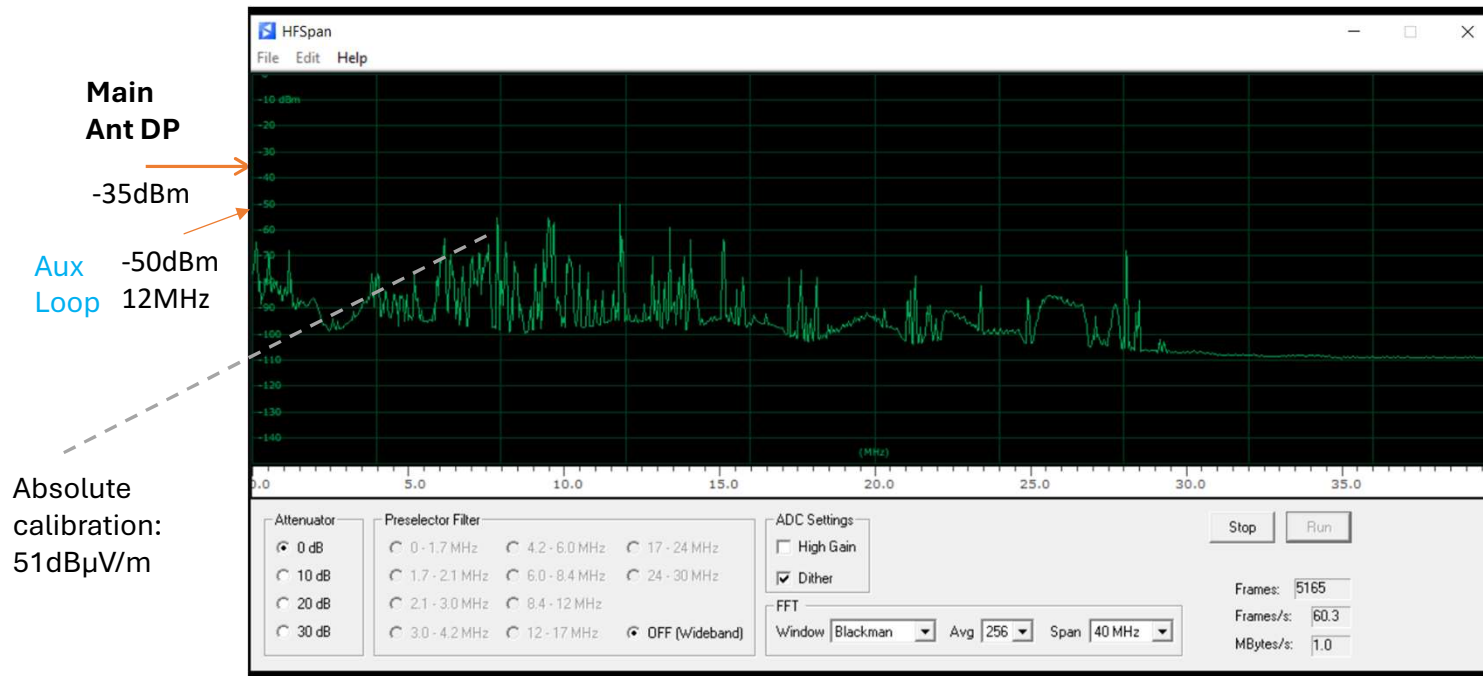
→
-35dBm



5. EMC case study HB9CVQ (12)

Signal levels (RX aux active mag Loop, 1m diameter, location attic)

13-Jan-2026-1650CET-horiz-ALA-1530 mag. Loop -55cm to tile-roof-250cm above concrete floor-Avg256

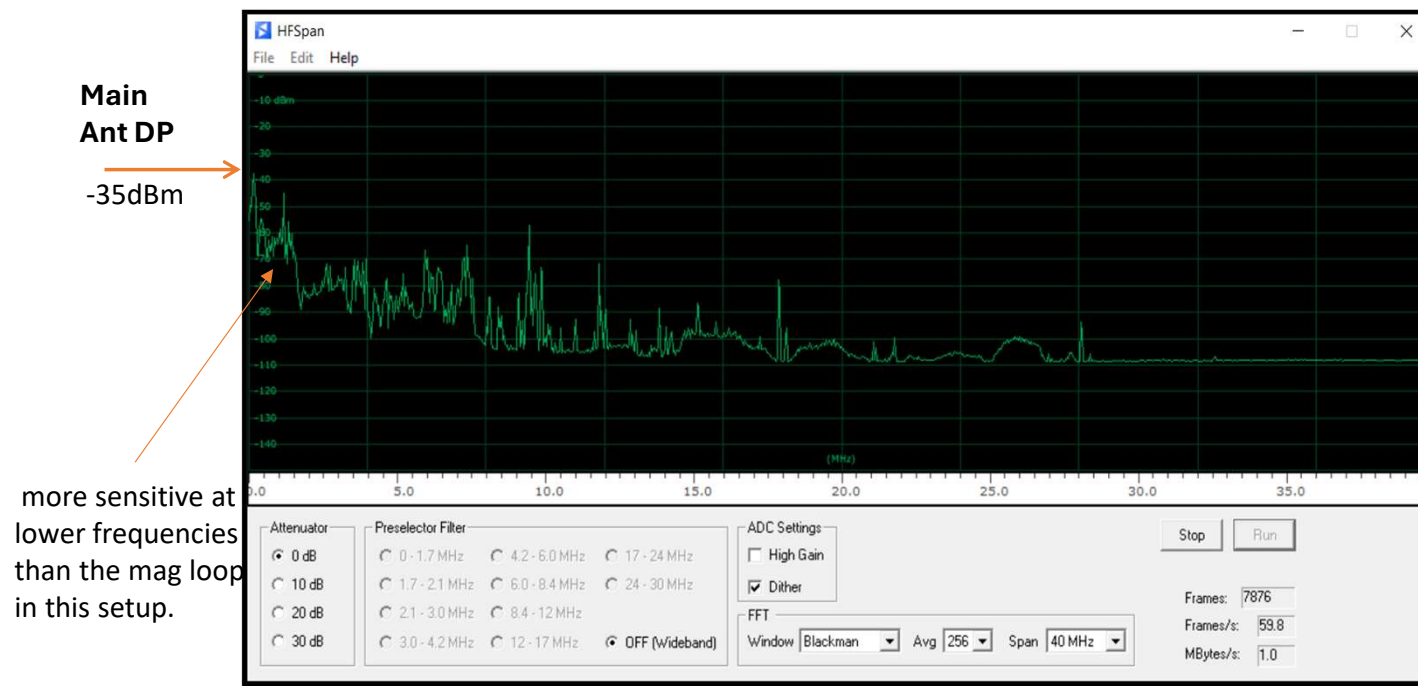


Around 17:44 CET, propagation checks showed nearly equal signals.

5. EMC case study HB9CVQ (11)

Signal levels (aux Mini-Whip PA0DRT, same location as before, attics)

13-Jan-2026-1731CET-Mini-Whip-with CM-Choke-150cm above concrete floor-Avg256



6. EMC case study Multi-OP CW contest station DD1A, Germany, Singen (160–10 m, ~750 W) (1)

- Multiple antenna masts; **close antenna coupling** (40 m @26m 2 el.; up to 4 el. beams; 20 to 10m stacks; 80 m GP; 80 m inv-V; 160 m L-antenna; home-built multi-in-band automation; multiple RX loops; beverages)
- BW mast, Versa Tower, old Armed Forces mast — critical direction-dependent antenna coupling.
- TRX, PA filters, 3-phase mains • **3 operating positions** — each on a single mains phase!
- **Simultaneous TX on different HF bands** (band-pass filters at TRX and after PA)
- Beverage RX antennas; ultra-narrow, high-IP BPFs (e.g., 3 positions for 80 m CW).
- One complete station per mains phase; mains filters; lots of large 300 g CM snap ferrites (DX-Wire K25, 25 mm ID).
- Almost consistent single-point entry EMC concept (no reference ground plane yet, connections to below club station DK0SV !).
- After HF, mains, CM-ferrite choking, we still found the individual TS-590 13.8 V DC supplies to be a critical back-door RX coupling path (via PE/SL/coax shields). Remedy: stronger DC-CM filtering without saturation and larger K24 cores (only 3–4 turns; more turns → capacitive crosstalk on core turn to turn → reduced choking function).

6. EMC case study Multi-OP CW contest station DD1A (2)

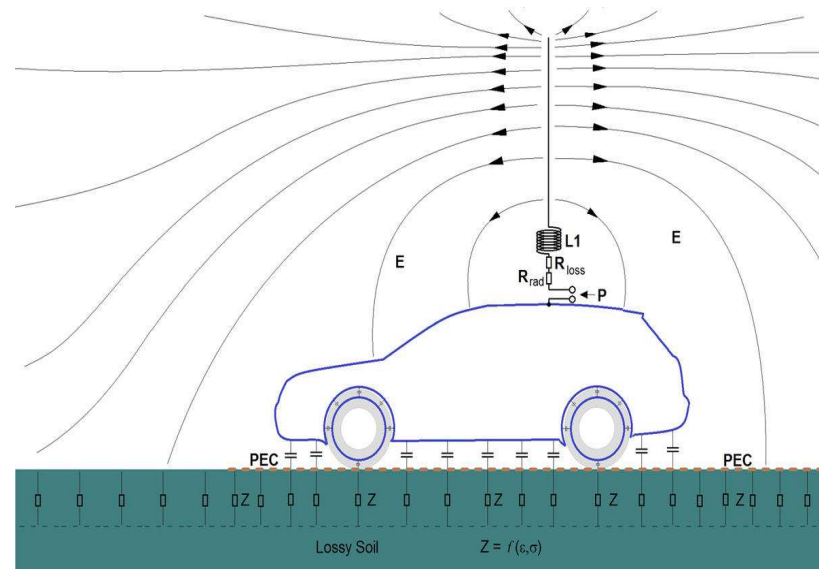
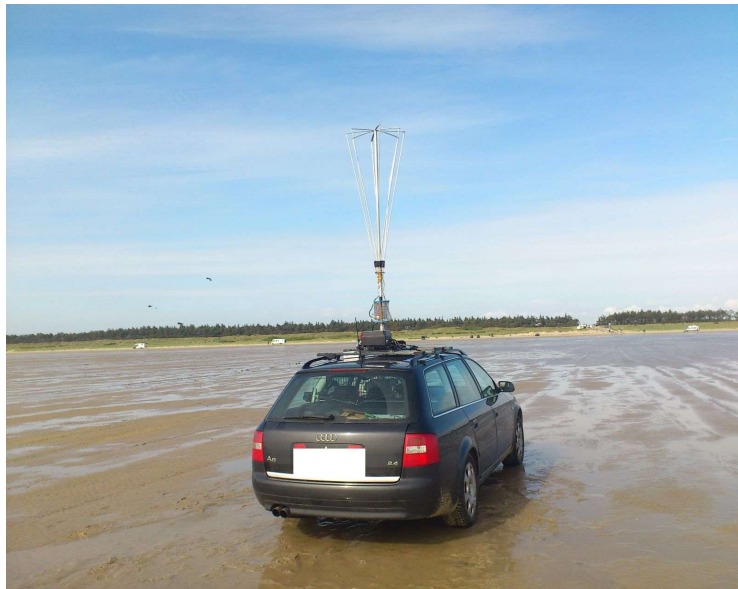
- The last **mitigation** weekend Jan-2026 was complex and time-consuming.
- Core **property size** roughly **40x25 m** with 3 ant. masts, + external 80GP/160m L antenna, all RX-Antennas
- On 3 mains phases run 3 filtered HF stations.
- Consistently, all (antenna) TRX/PA ports were suppressed with coax CM chokes and band-pass filters (automation, multi in-band !), on all 3 operating positions.
- These BPF filters typically give 30–40 dB isolation to the adjacent band-usually sufficient to push TX **harmonic signals** down, if no further uncontrolled EMI coupling exists.
- **TX**=> **1 kW** (+60 dBm) - 60 dB (2 filters) -20dB (stubs) -10dB to 20 dB Antenna decoupling = -30dBm (~**S9+40dB**)
- **RX**=> ideally should sees -30 dBm -80dB (2x BPF, 1x Stubs) = **-120dBm** (S1).
- Reality mostly at DD1A: **S9+20 (-53dBm)** on harmonics and +/-5 to 10kHz around ☹️ . It was even worse in the past.
- **Fixing** by better, stiffer grounding system and better isolation to club-station below using partly same antennas
- At least **additional** (target **S7: -85dBm**) **32 dB suppression needed** by e.g., improved ground plane/single point entry
- **Short circuited $\lambda/4$ Stubs add to CM pollution!**
- Antenna isolation can not be change due to property. Poor TX phase noise can principally also play a role.
- **Isolation/improvement of cross-talk needed mostly for RUN 1 to MULT-Operator in MS/M2 contests.**

6. EMC case study Multi-OP CW contest station DD1A (3)

-EMI-Mitigation-

- A key finding: **RX back-door** coupling via the DC supplies and **loops (PE/SL/coax shields)**.
- **TX field strength (self-illumination) induces disturbance currents.**
- The initial 20 dB **mains-filter spec** at external DC supplies was insufficient. Worse, at first two TRXs shared one 13.8 V DC supply and coupled uncontrollably; separating and choking yielded an acceptable solution.
- **Passive** (active needs BPF) **RX antennas** were now also better EMI suppressed by 40dB CM chokes
- All 3 Op-Position (TRX USB, PTT, CAT, etc.) were choked and mains filters with PE suppression installed.
- **Antennas are too close together** (limited property); directional antennas couple direction-dependently (USA/Asia).
- External lightning protection on the 24 m Bundeswehr mast is installed.
- Despite TRX-BPF → PA-BPF → stubs → antennas, we still see heavy crosstalk (see slide 48)

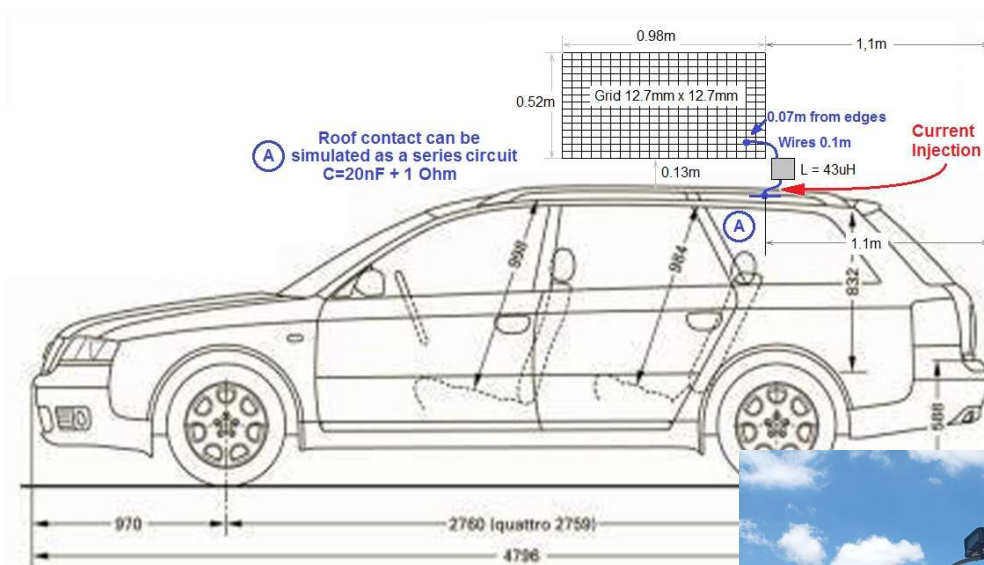
7. HF mobile challenge — Audi A6, HB9CVQ/m, DK2VQ/m (1)



7. HF mobile challenge — Audi A6, HB9CVQ/m, DK2VQ/m (2)

- A self-developed **80 m mobile antenna** proved very efficient under the circumstances by minimizing stray capacitance to the roof.
- Vehicle **EMI self-suppression** on an Audi A6 (2003) reduced initial S9++ (CW BW 400 Hz) to S3 with considerable effort.
- EMC-wise, the **passenger compartment** (K3S CW-only, 50 W HF) was treated as a 'Faraday cage'. Shielding to the outside measured ≈ -30 dB on 40 m, location-dependent (worst near steering wheel).
- The **RG-58** U feedthrough to the **antenna was bonded to chassis** right at the **passenger-door entry (zone concept)**.
- At the roof, a ring core choke was placed at the feed. There is an E-field gradient of ≈ 20 dB between floor pan and roof. **Strict single-point grounding inside avoids loops.**
- A **separate 90 Ah radio battery** was HF-decoupled and charged from the alternator via a DC-DC converter (constant 13.8 V).
- External AC power was never needed during 50 W operation across 30k+ km.

7. HF mobile challenge — Audi A6, HB9CVQ/m, DK2VQ/m (3)



various low bands antenna types tested



7. HF mobile challenge — Audi A6 (4)

- Various antenna prototypes were tested; efficiencies varied widely.
- Magnetic loops were particularly inefficient.
- **Radiator length is critical but limited by traffic regulations** (max 4 m from ground to tip). If the roof is ~1.5 m high, little remains.
- **Ground losses** on 80 m (night) and even 40 m (day) are **unfavorable**.
- On **160 m**, both technically and operationally, it's **practically hopeless** (QRPP).
- The **EMC zone-protection concept worked reliably across all R&D prototypes**.
- One might consider **bonding the car to radials** or even a ground rod when stationary - testing showed little to **no benefit**. This approach is even less feasible in aircraft in flight; you would need an impossible kilometers-long ground lead. (Note: VLF aircraft-to-submarines do use km-long trailing antennas.)
- **EMC-wise we do HF potential equalization referenced to any metal fuselage.**

7. HF mobile challenge — Audi A6 (5)

- Vehicle suppression (gasoline engine; in 2003 Audi had limited EMC measures): initially up to S9+20 dB of noise
- Relatively short (vs. wave-length) **low-band mobile antennas on the roof** (best place)
- Autonomous 12 V charger for radio battery
- QRV CW only on low bands, max 50 W HF output
- Due to **low antenna efficiency** $\Rightarrow$ effectively QRP

(e.g., 160 m $\approx$ 0.1%, 80 m $\approx$ 3%, 40 m $\approx$ 10% — 50 W $\rightarrow$ $\sim$ 5 W radiated)

- + Vehicle self-suppression done with **HF current clamp**; multiple bonding straps on engine, exhaust, etc.
- + **Multiple ground points** in the vehicle create loops; the main 12 V battery negative is on chassis; create a central bond near passenger door where coax shield, TRX K3S chassis, and antenna ground (via metal roof rack to roof rails) are finally brought to the car chassis.
- + **EMC disturbances depend on driving state and external environment** (QTH). On 40 m CW (400 Hz BW) at 120 km/h on an open highway (K3S/P3, optimized ATAS-120S), the S-meter dropped to S3.
- + **Classic vehicle EMC issues**: ground corrosion, mechanical vibration, temperature; height limit <4 m tip-to-ground; strong wind forces; other vehicles' EMC ($\gg$ V/m fields).
- + Potential administrative problems: Operating authorization while driving, insurance/liability, etc.

8. Summary and optimized practical recommendations (1)

- **Method: lines before fields; find root causes/sources** — think system-level EMC! **HF-island approach.**
- **Success is possible if you know what you're doing and have battery-powered, trustworthy instruments.**
- **Implement the EMC zone-protection concept systematically;** create a shack HF island.
- **Avoid/minimize coupling-loop areas; provide HF reference ground planes.**
- High suppression efficiency (for E, H, I, V) is often challenging at first.
- **Attenuation for current/voltages E-, H-Fields: $[\text{dB}] = 20 \cdot \log_{10}(\text{V}_{\text{ref}} / \text{V} \text{ e.g. choke})$;**
- **Attenuation for power: $[\text{dB}] = 10 \cdot \log_{10}(\text{V}_{\text{ref}} / \text{V} \text{ e.g. choke})$;**
- **RX** case: the environment (EM emissions) creates conducted coupling/disturbance currents (dB μ V/m, dB μ A).
- **TX** case: your own antenna creates the coupling field (V/m, mA).
- **Optimized EMC with good cost/benefit for both TX and RX is achievable.**
- *Don't think of ground rods first! 'Long' ground leads and lightning protection are useless for EMC.*
- Long water/heating pipe systems often act as disturbance-collecting antennas.
- Desk ground-plate, short bonding straps, mains filters with surge protection, ferrites, etc. ☺

8. Summary and optimized practical recommendations (2)

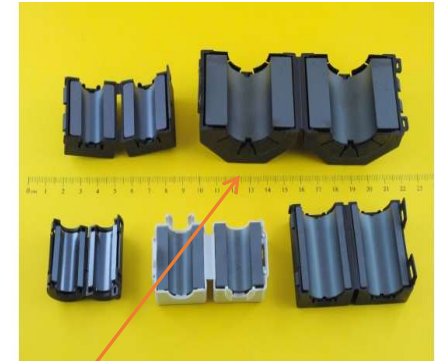
- Typical example of **cost/benefit for suppression** in both TX and RX — **affordable**.
 - **Systematic EMC design**: desk ground-plate; short studs/straps; single-phase mains filters with surge protection; common-mode chokes; snap ferrites/ring cores — component selection, etc.
- Useful vendors: <https://auth-nt.de> <https://www.dx-wire.de>



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- diverse Anschlußkabel optional

???



Tip: Fair-Rite part no. #0444177081,
~25 mm ID, useful ~1 MHz → ~500 MHz.
On HF, 3–4 turns usually suffice for ~20 dB.

HAM Radio YLs, OMs

Thank you all for your Time

*Understanding is the first step to find your
own, specific EM-Problem Solution, effectively!*

Any Questions / Comments / Discussion

Call us, **we help also professionally** since over 30+ years

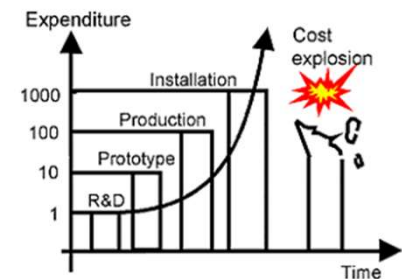
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